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PIHEM LIGHT

Passenger car and Heavy duty Emission Model

User Guide for Version 1
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1 Abbreviations and Symbols

Blue underlined symbols are input data for PHEMLight

Λ_v	Rotating mass factor calculated by linear interpolation from a polygon $\Lambda = f(v)^1$ [-]
a	acceleration of the vehicle in $[m/s^2]$
<u>A_{cross}</u>	cross-sectional area of the vehicle in $[m^2]$
<u>C_w</u>	drag coefficient [-]
<u>Fr_0, Fr_4</u>	rolling resistance coefficients
g	acceleration of gravity $[m/s^2]$
<u>Gradient</u>	road gradient in [%]
<u>m_{load}</u>	mass of the payload or passengers in [kg]
<u>$m_{rot,wheels}$</u>	Wheels equivalent rotational inertia [kg]
<u>$m_{vehicle}$</u>	vehicle curb weight in [kg]
P_{norm}	P_{wheel}/P_{rated} for HDV and for FC of all vehicles [-] P_{wheel}/P_{drive} for regulated emissions of PC and LDV [-]
P_0	ratio of power consumption of auxiliaries to the rated power of the internal combustion engine [-]
P_a	acceleration power in [kW]
$P_{a,max}$	maximum power available for acceleration [kW]
P_{air}	power demand from air drag [kW]
P_{aux}	power demand from auxiliary units [kW]
P_{drive}	wheel power at 70 km/h speed and $0.45 m/s^2$ [kW]
P_e	effective engine power output at crankshaft [kW]
P_{grd}	power demand from road gradient [kW]
$P_{max,norm}$	available share of rated power calculated by linear interpolation from a polygon $P_{max,norm} = f(v)$
<u>P_{rated}</u>	rated engine power [kW]
P_{roll}	power demand from rolling resistance [kW]
P_{transm}	transmission losses [kW]
<u>v</u>	vehicle speed [m/s] (converted from [km/h] in the *.dri file)
η	drive train efficiency (0.8 for HDV_CB and 0.9 for all other)
ρ	air density $[1.2 kg/m^3]$

¹ In PHEM light the rotating mass factor is defined slightly different from literature and comprises only the rotational inertias from engine and drivetrain (but not of the wheels)

2 Introduction

The PHEMLight model is a simplification of the model PHEM (Passenger Car and Hheavy Duty Emission Model) to be integrated into traffic models (e.g. SUMO²). It overcomes the following problems of PHEM by the combination with a traffic model:

- Information's that are needed several time steps before and after the actual computed time, which PHEM needs to calculate the thermal status of after-treatment systems, to consider effects of transient engine loads on the engine out emissions and to feed the driver model which selects the gears in the virtual vehicles
- High storage demand and long computation time, because PHEM needs to hold long time series for each vehicle in the simulator

These problems were overcome by replacing the needed time steps before and after the actual computed time by generic functions which use only information's that are available for the actual second.

PHEMLight calculates fuel consumption and pollutant emissions of vehicles on the basis of vehicle data to be entered for various driving cycles on basis of characteristic emission curves and vehicle longitudinal dynamics. Since the vehicle longitudinal dynamics model calculates the engine power output and speed from physical interrelationships, any imaginable driving condition can be illustrated by this approach. The simulation of different payloads of vehicles in combination with road longitudinal gradients and variable speeds and accelerations can thus be illustrated by the model.

PHEMLight offers a very extensive database of previously measured vehicles and engines for the calculation of traffic emissions in road networks. This data is compiled as average vehicles, which illustrates the vehicle categories of passenger, light duty and heavy duty vehicles with Otto and diesel engines from EURO 0 to EURO 6c for each engine. Also provided are data for hybrid and battery electric vehicles. The user needs to enter either only the speed curves.

The following calculation modes are available:

- PHEMLight STANDARD: Simulation of individual vehicles in defined speed curves.
- PHEMLight BATCH: Addition to PHEMLight STANDARD, which enables batch processing of vehicle and driving cycle files.

The database for the average vehicles in PHEMLight is identically to PHEM. The emission curves were generated with PHEM.

3 Function Overview

The PHEMLight model calculates the necessary engine power output for the actual second for a given driving cycle (speed curve and road longitudinal gradient over time) from the driving resistances and drivetrain losses. The emissions are then interpolated from emission curves in the normalised format according to the current engine power output.

²<http://sumo-sim.org/>

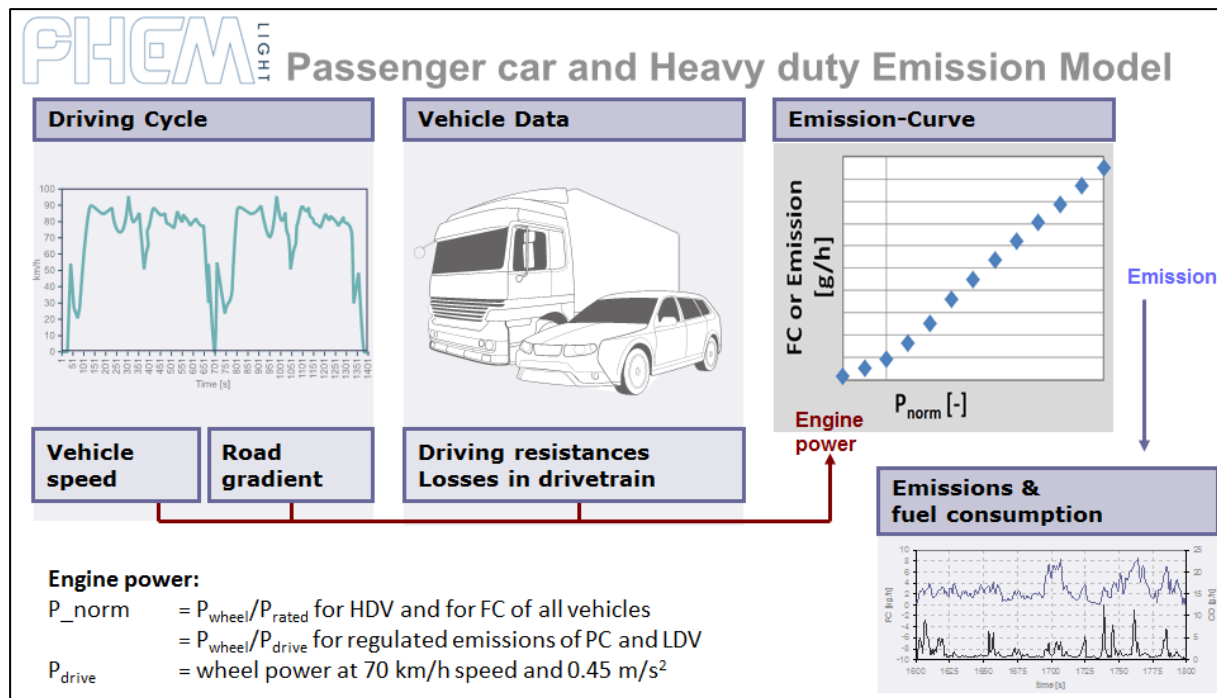


Figure 1: Schematic picture of the PHEMLight model

The consumption and emission values are normalised

- for heavy duty vehicles by dividing by the engine rated power $[g/h)/kW_{rated}]$
- for passenger and light duty vehicles for the fuel consumption also by dividing by engine rated power. The pollutant emissions are normalised by the driven engine power at 70km/h speed and 0.45m/s^2 .³

The normalised emission curves format makes it possible to create average curves. In PHEMLight this can be done over the FleetMix option where the users can define an average vehicle for the classes Passenger Cars (PC), Heavy duty vehicles (HDV), Busses (BUS) and Two-wheelers (TW).

Apart from the emission curves, all of the vehicle data required for calculating the engine power output has to be predefined for the model (weight of the vehicle and load, drag coefficient, cross-sectional area, rolling resistance coefficients, final drive and transmission ratios etc.). This database is currently available for the vehicle categories shown in Table 4 and was parameterised by the PHEM vehicle database. Like in PHEM, the different vehicle size classes are currently only enabled for light duty and freight vehicles. For all the other vehicle categories, there is only an “average size”.

³ The normalisation applied for pollutant emissions at cars and LCVs considers the effect that regulated emissions are lower in load areas controlled by type approval than in outer areas. The load area controlled by type approval testing of vehicles is defined by a speed and acceleration cycle. Thus the normalisation by a power defined by representative speed and acceleration values is better than a normalisation with the engine rated power.

4 PHEMlight vehicle naming logics

The standard vehicles in PHEMlight are named using combination of the vehicle codes shown in Table 1. Explanation for the codes can be found in Table 2 and Table 3.

Table 1: PHEMlight vehicle description codes

	optional		optional
Vehicle category *1	Size class *2	Technology	Emission standard *3
PC	I	G	EU0
LCV	II	D	EU1
HDV_RT	III	G_HEV	EU2
HDV_TT		D_HEV	EU3
HDV_CB		CNG	EU4
HDV_CO		BEV	EU5
MC_2S			EU6
MC_4S			EU6c
MOP			

*1: Work on dataset for 2-wheelers ongoing

*1: Sizes classes used for LCVs (I, II, III) and HDV_RT (I, II)

*2: Emission standard not applicable for BEV

Examples:

PC_G_EU6 passenger car, gasoline engine, EURO 6
PC_BEV passenger car, battery electric
 light commercial vehicle, size class III, CNG engine, Euro
LCV_III_CNG_EU5 5
HDV_RT_I_D_EU4 rigid truck, size class I, diesel engine, EURO IV
HDV_CB_D_EU5 city bus, diesel engine, Euro V

Table 2: Explanations vehicle category and size class

code	explanation	
PC	passenger cars	
LCV	light commercial vehicles; differentiation into size classes according to legislation (N1-I, N1-II, N1-III)	
	size class I	RM < 1305kg e.g. VW Polo CityVan, Ford FiestaVan ...
	size class II	1305kg < RM <= 1760kg e.g. VW Caddy, Renault Kangoo
	size class III	1760kg < RM <= 3500kg e.g. Iveco Daily
	"RM": reference mass (approx. vehicle curb weight)	
HDV_RT	heavy duty vehicles - rigid trucks	
	size class I	2 axle trucks
	size class II	3+ axle trucks
HDV_TT	heavy duty vehicles - truck + trailer (incl. articulated trucks = tractor + semitrailer)	

HDV_CB	heavy duty vehicles - city bus
HDV_CO	heavy duty vehicles - coach
MC_2S	motorcycles - 2 stroke
MC_4S	motorcycles - 4 stroke
MOP	moped

Table 3: Explanations technology code

Explanations: technology

G	gasoline engine (conventional powertrain)
D	diesel engine (conventional powertrain)
G_HEV	gasoline engine (parallel hybrid powertrain)
D_HEV	diesel engine (parallel hybrid powertrain)
CNG	CNG engine
BEV	battery electric vehicles

Remarks on hybrid electric vehicles (HEVs):

The fleet average "HEV" vehicles in PHEM and PHEMLight do NOT match with single particular vehicles in the market (e.g. Toyota Prius) but have been defined as follows:

- 1) Baseline vehicle setup (e.g. vehicle size) similar than related conventional vehicle
- 2) Total rated vehicle power of baseline conventional vehicle split to combustion engine and to electric motor (EM)
- 3) Additional vehicle weight due to EM and battery considered
- 4) Application of generic parallel hybrid operation strategy (recuperation, boosting, charging etc.)

Table 4 gives an overview on all available vehicles in PHEMLight.

Table 4: Available vehicles in PHEMLight

	Otto engine		Diesel engine		BEV	CNG
	<i>Conventional</i>	<i>Hybrid</i>	<i>Conventional</i>	<i>Hybrid</i>	<i>Conv.</i>	<i>Conv.</i>
PC	EU0-EU6c	EU5-EU6	EU0-EU6c	EU6	available	EU5-EU6
LCV	EU0-EU6c (I-III)	EU6 (I-III)	EU0-EU6c (I-III)	EU6 (I-III)	EU6 (I-III)	EU6 (I-III)
HDV_RT	n.a.	n.a.	EU0-EU6 (I-II)	n.a.	n.a.	n.a.
HDV_TT	n.a.	n.a.	EU0-EU6	n.a.	n.a.	n.a.
HDV_CO	n.a.	n.a.	EU0-EU6	n.a.	n.a.	n.a.
HDV_CB	n.a.	n.a.	EU0-EU6	EU6	available	EU6

All the input data is saved in separate files and added to the program in the PHEMLight package.

5 Installation

PHEMLight is a portable application which does not require running a setup. Just copy the application in a directory with read and write access. The "Default Vehicles" folder must be copied in the same directory for use.

5.1 First time application start & licensing

For PHEMLight no license is needed. But the "Default Vehicle" data is not in total free. Only the vehicle data for PC-EU4 for Otto and diesel engines is free to use. All other vehicle categories must be bought.

On first startup PHEMLight will create subfolders and various configuration files.

5.2 System Requirements

PHEMLight should be able to run on any system that has **Microsoft .NET Framework 4.0 or higher** installed. The following format is used for all input and output files:

- **Comma (,) as list separator**
- **Full stop (.) as decimal separator**

Make sure you use this format when opening and saving input files.

6 User Interface

The user interface documentation is available in PDF and can be loaded directly from the application using the  button. Alternatively you can find the PDF user manual in the "Documentation" folder.

6.1 PHEMLight dll

The PHEMLight calculation method is available as dll. This dll is programmed in C# and can be included in most programs.

7 Calculation Modes

7.1 STANDARD Mode

This is the default calculation mode and allows one or more GEN files to be run independently from each other. The files need to be loaded in the job file list with activated checkboxes. Modal results will be written for each GEN file. The summary file (.erg) includes the results of all calculations.

7.2 BATCH Mode

Switch to BATCH Mode using the mode selection in the Options tab. In this mode a list of GEN files (job list) and a list of DRI files (cycle list) is defined in the corresponding tabs. During calculation every GEN file (= vehicle) will be calculated with every DRI file (= cycle) defined in the lists (only checked items!). **The driving cycles defined in the GEN files will be ignored!**

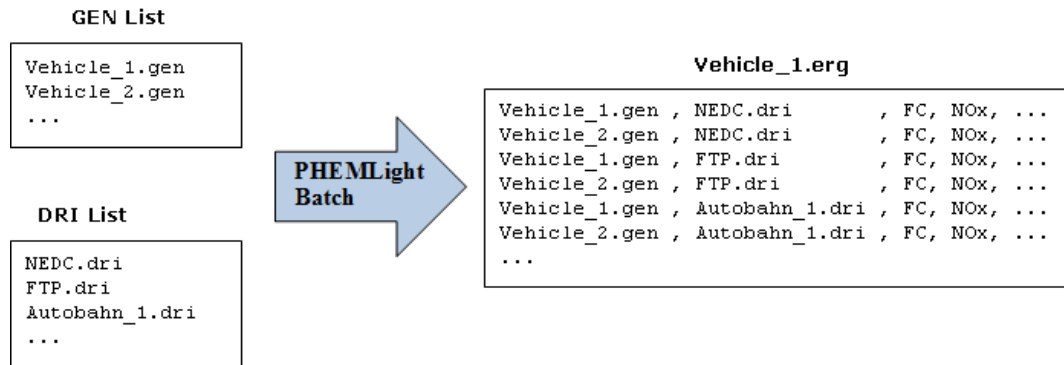


Figure 2: PHEMLight BATCH mode

The average results for every calculation are output in an .erg file. The .erg file is named after the first file in the GEN list and is stored in the same folder.

8 In-/output files

Prior to running the program, the required input files have to be created or the previously created input files have to be selected. All input files are saved in a file with a “gen” extension.

If all required input files are available, PHEMLight starts the calculation and writes the result files into the path which is defined in the “.gen” file, except in Batch Mode which allows a custom output path to be defined in the Options Tab. The result files have the same file name as the “.gen” file, but with a different extension, see below.

8.1 Job files

Job files include all settings and paths to input files required for calculation.

.genJob file for STANDARD and BATCH mode. These files can be created and modified with the GEN Editor.

8.2 Input Files

.veh Vehicle and engine specifications.

_FC.csv.....Fuel consumption curve

.csv.....Emission curve.

.driDriving cycle.

File templates are available in the "Documentation\Input Files" folder.

8.3 Output Files

STANDARD/BATCH Mode:

“.erg”summary/cycle-averaged results for the calculated vehicles

“.sta”second-by-second results for the calculated vehicle

Descriptions of the output parameters can be found in the "Documentation\Result Files" folder.

Note that PHEMLight overwrites existing files without warning!

9 Calculation Process

9.1 Longitudinal dynamics and emission calculation

The most important calculation steps are described as follows:

For the most accurate calculation of power output, the most important parts of power output under real operating conditions of a vehicle are taken into consideration. In detail, these are rolling and air resistance, acceleration and road gradient, transmission losses as well as power demand of auxiliaries, unless they have already been considered in the effective engine power output. Tire slip is not taken into consideration in the present version (little influence on normal operation and currently no generally valid approach in PHEMLight). The approach is as follows:

Equation 1: Formula for calculating the driving power in the cycle:

$$P_e = (P_{roll} + P_{air} + P_a + P_{grd})/\eta + P_{aux}$$

With η from 0.9 for all vehicle classes excluded HDC_CB which have an efficiency from 0.8.

The calculation of the individual parts of power output is described as follows:

Equation 2: Power to overcome rolling resistance in [W]

$$P_{roll} = (m_{Vehicle} + m_{Load}) \times g \times (Fr_0 + Fr_1 \times v + Fr_4 \times v^4) \times v$$

Equation 3: Power to overcome air resistance in [W]

$$P_{air} = C_w \times A_{cross} \times \frac{\rho}{2} \times v^3$$

Equation 4: Acceleration power in [W]

$$P_a = (m_{Vehicle} \times \Lambda_v + m_{rot, wheels} + m_{Load}) \times a \times v$$

The masses of vehicles and payloads are accelerated translationally and all rotating components of the vehicle (tires etc.) are accelerated rotationally. The rotationally accelerated masses are categorised in three groups. Each of these has a different transmission ratio and thus

goes through different angular acceleration. The rotational acceleration power output for these three component groups is converted into translational acceleration:

Equation 5: Calculation of equivalent rotational masses

The equivalent rotational masses are interpolated from curves specified in the vehicle file dependent of the vehicle speed. The curves are generated for each possible gear number and vehicle class. The speed curve comes from the average velocity for each gear in the ERMES cycle. The rotating mass factor (Λ_v) curve was calculated over the average values for each gear from a full load acceleration.

$$\Lambda_{v,i} = \frac{\frac{P_{a,max} \times 1000}{a_{max} \times v} - m_{Loading} - m_{rot,wheels}}{m_{Vehicle}}$$

With:

$$P_{a,max} = P_{rated} \times P_{max,norm} - P_{aux} - P_{transm} - P_{roll} - P_{air} - P_{grd}$$

Equation 6: Power to overcome the road gradient in [W]

$$P_{grd} = (m_{Vehicle} + m_{Load}) \times Gradient \times 0.01 \times v$$

Where:

[Gradient](#)road gradient in [%], entered in the *.dri file

Equation 7: Power consumption of auxiliaries [W]

$$P_{aux} = P_0 \times P_{rated}$$

Where:

P_0 ratio of power consumption of auxiliaries to the rated power of the internal combustion engine [-]

P_{rated} rated power of the internal combustion engine

Most of the time, the accurate capture of the power demand of auxiliaries (e.g. air-conditioning) under real operating conditions is very costly and time-consuming. If a more detailed graph is needed, please contact the FVT.

9.1.1 Background of emission curves (_FC.csv, .csv)

The specific emission curves have been elaborated by the software PHEM. Based on a standard driving cycle PHEM calculates the emissions and the corresponding wheel power value in 1 Hz. The simulated emissions are rasterised over normalised engine power areas into a characteristic curve.

Table 5: Cycles for characteristic curve generation

Vehicle category	Cycle
PC	ERMES_V7
LCV	ERMES_V7
HDV_RT	TNO_HD_12kWpt
HDV_TT	TNO_HD_12kWpt
HDV_CO	Coach_HDV_CO2
HDV_CB	Citybus_HDV_CO2

9.1.1.1 Fuel consumption curve (_FC.csv)

In the specific fuel consumption curve, the fuel consumption is stored over the normalised engine power output. For the consideration of idling values, the idle value is also stored at the first position.

Units:

- Fuel consumption in [(g/h) per kW rated power]

9.1.1.2 Emission curve (.csv)

In the specific emission curve, the emissions for NO_x, HC, CO, PM, PN and NO are stored over the normalised engine power output by HDV vehicles and for PC and LCV over the engine power normalised over the driven engine power at 70km/h speed and 0.45m/s². For the consideration of idling values, the idle values are also stored at the first position.

Units:

- Emissions in [(g/h) per kW drive power] for PC and LCV
- Emissions in [(g/h) per kW rated power] for HDV

9.1.1.3 Driving Cycle (.dri)

When calculating vehicles the vehicle speed is the only required parameter. Additionally parameters can be given for road gradient and time. If both are not given, PHEMLight uses time steps from one second and a road gradient from 0%.

Figure 3: Driving cycle example

```
v1,,
<t>,<v>,<grad>
[s],[km/h],[%]
1,33.95,0
2,34.9,0
3,35.5,0
4,35.5,0
5,36.1,0
6,37.4,0
7,38.8,0
8,40.275,0
9,41.825,0
10,43.409,0
11,44.975,0
12,46.503,0
13,48.042,0
14,49.574,0
15,51.041,0
16,52.465,0
17,53.893,0
18,55.256,0
19,56.552,0
20,57.818,0
21,59.057,0
22,60.261,0
23,61.432,0
24,62.521,0
25,63.531,0
26,64.508,0
27,65.378,0
```

9.2 Fleetmix Mode

For the use of fleet mixes an FleetShares.csv file is needed. In this file the user can defined his average vehicle for PC (PC and LCV), HDV (HDV_RT and HDV_TT), BUS (HDV_CO and HDV_CB) and TW (MC_2S and MC_4S). In each of these categories the user can give the vehicle class and the share on the total which must be one. An example of such FleetShare.csv file can be seen in Figure 4.

In the folder where the FleetShares.csv file is stored, all the vehicle data must also be stored.

Figure 4: FleetShare File for Fleetmix mode

```
c Fleet composition for link of PHEMLight with NITRA @,,
c,,
c fleet segment,vehicle segment,share
PC,PC_D_EU0,0.000561356
PC,PC_D_EU1,0.007212314
PC,PC_D_EU2,0.011723856
PC,PC_D_EU3,0.024710612
PC,PC_D_EU4,0.07385672
PC,PC_D_EU5,0.202361042
PC,PC_D_EU6,0.262347759
PC,PC_G_EU0,0.00443611
PC,PC_G_EU1,0.010251764
PC,PC_G_EU2,0.009507202
PC,PC_G_EU3,0.006406532
PC,PC_G_EU4,0.055329154
PC,PC_G_EU5,0.142955882
PC,PC_G_EU6,0.188339696
HDV,HDV_RT_D_EU5_I,0.2
HDV,HDV_RT_D_EU5_II,0.2
HDV,HDV_RT_D_EU6_I,0.3
HDV,HDV_RT_D_EU6_II,0.3
```